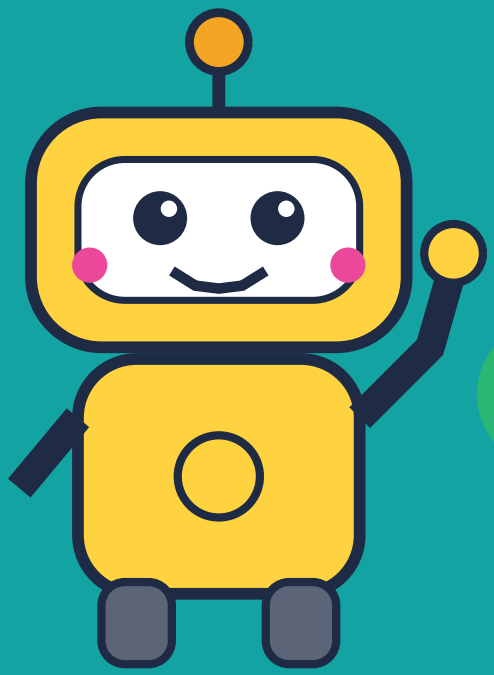
SPECIMEN BOOKLET

Think Like a Coder

Computational thinking for Classes 3–5

A sample for schools: three full chapters, a teacher script, the classroom kit and how assessment works



For CBSE's Computational Thinking and AI curriculum, 2026-27

Class 4 sample · Follows the NCERT maths chapters · No computers needed

Practice Pack Rs 449 per child (indicative) · Free pilot, January-March 2027

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CodeSchool is a working name. Not an official CBSE or NCERT publication.

A complete CT&AI programme for Classes 3-5

From 2026-27, CBSE asks schools to teach Computational Thinking and AI in Classes 3-8: **50 hours a year in Classes 3-5**, built into Mathematics and The World Around Us periods. CBSE's free handbooks give puzzles for each maths chapter. CodeSchool adds everything a teacher needs to teach it well, with no computers needed.

What schools need	CBSE's free handbook	CodeSchool Practice Pack
Puzzles for each maths chapter	Yes	Yes, in levels from easy to challenge
Theory in simple words	Short introduction	A Learn page and Robo's tip in every chapter
Worked examples	No	One per chapter, step by step
Coding	No (follow steps only)	'Now you build it': coding on paper with code cards
Hands-on activities	One or two a year	A classroom kit used all year
Teacher lesson plans	Answer keys	A period-by-period teacher script book
Assessment and report card	No format given	Self-check, observation journal and term summary
Teacher training	CBSE workshops	School orientation, termly workshops and support

★ Why schools choose it

- ✓ Follows the chapters of the NCERT Class 4 maths textbook in the same order, so it fits the maths period.
- ✓ Built around the CT skills in CBSE's curriculum: steps, patterns, abstraction, decomposition, logic and data.
- ✓ Works in any classroom: paper, cards and a floor mat. A projector helps but is optional.
- ✓ Ready for a maths or EVS teacher, not only a computer specialist.

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What each school receives

For	What they get	Classes
Each child	Practice Pack: level workbook with coding cut-outs and take-home challenges	3, 4, 5
Each class section	Classroom kit: code cards, floor grid mat, cipher wheels, sorting cards, box	3, 4, 5
Each teacher	Teacher script book, answer keys, observation journal, short how-to videos	3, 4, 5
The school	Orientation at school, a workshop every term, a support line and a term summary format	All

The series

Book	Levels	Coder skills
Class 3 Practice Pack	B1-B3	Steps and codes, views, loops, decomposition, patterns, events, data, logic, sorting, if-then, variables, grids
Class 4 Practice Pack (this sample)	B3-B4	If-else choices, parity checks, sorting networks, nested and repeat-until loops, decision trees, data
Class 5 Practice Pack	B5-B7	Variables, search strategies, debugging and a game designed on paper
Classes 6-8 Projects and AI Pack	C1-C8	Functions, data, the AI chapters and two 20-hour projects a year (separate sample)

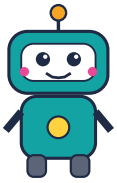
Price and ordering

Indicative price: Rs 449 per child per year (MRP) for the Classes 3-5 Practice Pack, through the school's book list. It can replace an existing computer book, so the cost to parents stays about the same. Kits, teacher books and training come with the packs. Prices are indicative until confirmed in writing.

Try it first

A **free pilot** in two or three sections from January to March 2027: materials, a teacher orientation, weekly support and a short report against success measures agreed in advance. No obligation to continue.

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Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn — Robo explains one big idea in simple words, with the new coder words to remember.

Example — A worked example shows the idea step by step.

Try it! — Five practice questions. Circle the answer or write it in the box.

Now you build it — Coding on paper: put code cards in order, fix bugs and write your own programs.

Challenge — A harder puzzle for when you are ready for more.

How did I do? — Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this | **Repeat: a loop** | **If: a decision** | **When: an event** | **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 4 maths textbook, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Inside the full Class 4 book

The complete book has 14 chapters, one for each chapter of the NCERT Class 4 maths textbook, plus answers, a glossary and cut-out code cards. This sample includes chapters 2, 4, 9 in full.

Ch	Chapter	Coder skill	Level	In this sample
1	Shapes Around Us	Faces, edges and nets	B1 review	
2	Hide and Seek	If-then-else choices	B3	Yes
3	Patterns Around Us	Odd, even and checking for mistakes	B3	
4	Thousands Around Us	Big numbers and sorting machines	B4	Yes
5	Sharing and Measuring	Fractions and checking a condition	B3	
6	Measuring Length	Shortest paths and loops with turns	B4	
7	The Cleanest Village	Tracing a machine step by step	B3	
8	Weigh It, Pour It	If-else decisions and measuring steps	B3	
9	Equal Groups	Loops inside loops	B4	Yes
10	Elephants, Tigers and Leopards	Clue tables and data	B3	
11	Fun with Symmetry	Mirror patterns and repeat with turns	B4	
12	Ticking Clocks and Turning Calendar	Repeat until and cycles	B4	
13	The Transport Museum	Decision trees: sorting by questions	B3	
14	Data Handling	Collecting and showing data	B2 review	

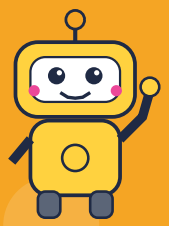
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CHAPTER 2 · LEVEL B3

2 Hide and Seek

Coder skill: If-then-else choices

Goes with NCERT maths, Class 4, chapter 2

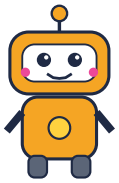


Learn

Computers make choices with **if-then-else**: *if* something is true, *then* do one thing, *else* do another.

When an if-else sits **inside a loop**, it is checked again every time the loop repeats. That is how a robot reacts to each new square.

if-then-else	condition
a choice between two actions	the question that is true or false



I check my if-else rule on every square: if it is a puddle I say 'Splash!', else I say 'Step'. Two choices, every time!

Example

Robo walks along the path

1. Program: *repeat 4 times: move forward; if on a puddle (grey) then say 'Splash!' else say 'Step'.*
2. Robo lands on tiles 1, 2, 3, 4: grey, white, grey, white.
3. Robo says: **Splash, Step, Splash, Step**.



★ *Coder fact!* A lift uses if-then rules: if someone presses a floor button, then it stops at that floor.

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Try it!

Circle the right answer, or write it in the box.

1 Use the same program on this path. What does Robo say?



2 Rule: *if the number is more than 10 then clap, else jump*. What do you do for 7, 12 and 10?

3 How many times does Robo say 'Splash!' on this path?

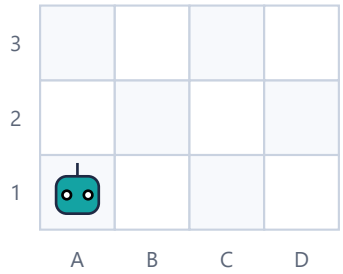


a) 2 **b)** 3 **c)** 4 **d)** 6

4 Robo is hiding in the red, blue or green box. *Robo is not in the red box. If Robo were in the green box, it would be open. The green box is closed.* Where is Robo?

a) red **b)** blue **c)** green **d)** can't tell

5 Robo is on A1 facing **up**. Program: *forward, turn right, forward, forward*. Which square does Robo reach?



a) C2 **b)** B3 **c)** A3 **d)** C1

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Now you build it

Write a program that makes Robo say 'Hot!' on sun tiles and 'Cold!' on snow tiles along a path of 5 tiles.

Repeat 5 times | move forward | If on a sun tile | say 'Hot!' | else | say 'Cold!'

Draw a path of 5 sun and snow tiles. Swap with a friend and trace each other's program.

Challenge

On a path of 10 tiles, every 3rd tile is a puddle. Robo runs the 'Splash or Step' program on all 10 tiles. How many times does it say 'Splash!'?

How did I do?

I can follow an if-then-else rule.	I can trace a program with a choice.	I can write my own if-else program.
		

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CHAPTER 4 · LEVEL B4

4 Thousands Around Us

Coder skill: Big numbers and sorting machines

Goes with NCERT maths, Class 4, chapter 4

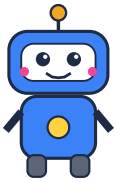


Learn

Big numbers have a **thousands** place: 4,072 means 4 thousands, 0 hundreds, 7 tens and 2 ones.

A **sorting network** sorts numbers with **comparators**. Two numbers enter a comparator; the smaller goes to the upper wire and the bigger to the lower wire. Computers can do many comparisons at the same time, so they sort very fast.

thousands the 4th place from the right	**comparator** compares two numbers and orders them	**sorting** putting in order

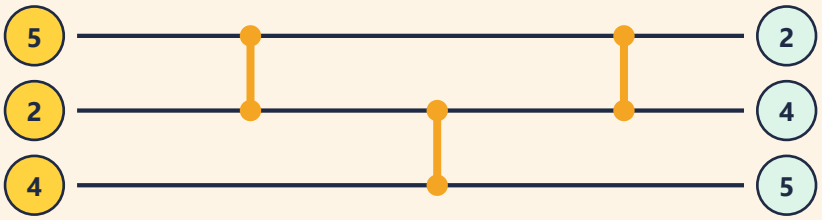


In a sorting network, numbers race along wires. At each comparator the smaller one goes up. At the end they come out in order!

Example

Sort 5, 2, 4 with the network

1. Comparator 1 (top two wires): 5 and 2, the smaller goes up: **2, 5, 4**.
2. Comparator 2 (bottom two wires): 5 and 4: **2, 4, 5**.
3. Comparator 3 (top two wires): 2 and 4 stay. Out come **2, 4, 5**, smallest on top.



★ *Coder fact!* Big websites sort millions of items in a blink by doing many comparisons at the same time.

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Try it!

Circle the right answer, or write it in the box.

1 Write 4,072 in parts: ___ thousands, ___ hundreds, ___ tens, ___ ones.

2 Order from smallest to largest: 3,410; 3,041; 3,401; 3,140.

3 7 and 3 enter a comparator. Which number comes out on the upper wire?



a) 7 **b)** 3

4 Send 9, 4, 6 through the same network as the example. What comes out, top to bottom?



5 Use the digits 5, 0, 8 and 1 once each. What is the largest number you can make?

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Now you build it

Turn your class into a sorting network for 3 friends of different heights.

Friends stand on wires 1, 2, 3

Compare wires 1 and 2: shorter goes to 1

Compare wires 2 and 3: shorter goes to 2

Compare wires 1 and 2 again

Try it with three friends. Do they always end up shortest to tallest?

Challenge

Could just the first two comparators always sort 3 numbers? Try 3, 2, 1.

How did I do?

I can read numbers in the thousands.	I can order big numbers.	I can use a sorting network.
		

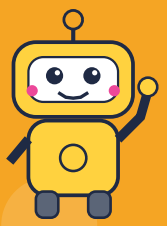
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CHAPTER 9 · LEVEL B4

9 Equal Groups

Coder skill: Loops inside loops

Goes with NCERT maths, Class 4, chapter 9

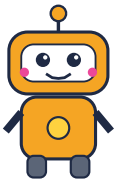


Learn

A **nested loop** is a loop inside another loop. The inner loop finishes all its repeats each time the outer loop runs once.

Nested loops make **arrays**: rows and columns. 3 rows of 4 is 3 x 4 = 12. Multiplication is a loop inside a loop!

nested loop	array	row
a loop inside a loop	objects in rows and columns	a line going across

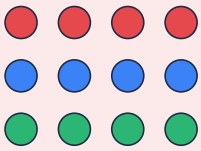


Repeat 3 times: [repeat 4 times: draw a star]. A loop inside a loop draws 3 rows of 4 stars: 12 stars!

Example

Stars in rows

1. Outer loop: repeat 3 times (one row each time).
2. Inner loop: repeat 4 times: draw a star. Then start a new row.
3. Total stars: 3 rows x 4 stars = **12**.



Coder fact! Screens are grids of tiny dots called pixels. A computer colours them row by row with loops inside loops.

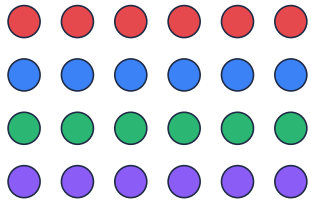
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Try it!

Circle the right answer, or write it in the box.

1 How many stars does this draw? *Repeat 5 times: [repeat 3 times: draw a star], new row.*

2 How many dots are in this array?



3 Which nested loop draws 18 stars?

a) repeat 3 x [repeat 6]
b) repeat 4 x [repeat 4]
c) repeat 5 x [repeat 3]
d) repeat 2 x [repeat 8]

4 There are 6 boxes with 5 pencils in each. How many pencils in all?

5 Draw what this makes: *repeat 2 times: [repeat 3 times: draw a circle], new row.*

Now you build it

Plan a garden with 4 rows and 5 plants in each row. Write it as a nested loop.

Repeat ___ times **Repeat ___ times** **plant a seed** **start a new row**

How many seeds do you need?

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★ Challenge

Repeat 3 times: [repeat 2 times: clap], stamp. How many claps and how many stamps?

✓ How did I do?

I can count with a nested loop.	I can read an array.	I can write a loop inside a loop.
		

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Sample teacher script: Robo makes choices

From the teacher book, Class 4, chapter 2 (*Hide and Seek*). One 40-minute period. Level B3.

You need	
Kit	Floor tiles or a strip of 5 squares (two marked as puddles), Robo card, if-else code cards
Book	Chapter 2 pages, 'Try it' questions 1-3
Goal	Children trace and write an if-else inside a loop, and explain why the check runs every time

Step and time	Teacher does	Children do
1. Predict and run (8 min)	Draw a row: START, puddle, plain, puddle, plain. Show: repeat 4 times: move forward; if on a puddle then say 'Splash!' else say 'Step'.	Write what Robo will say. One child acts as Robo while the class reads the program aloud (Splash, Step, Splash, Step).
2. Investigate (8 min)	Ask: 'What if the if-else were outside the loop? How many times would Robo check?'	Discuss in pairs and explain that the check happens every time the loop repeats.
3. Modify (8 min)	Draw a new row with a star square.	Change only the if-else so Robo says 'Star!' on star squares.
4. Put in order (8 min)	Hand out jumbled blocks, one of which is a trick card.	Put the blocks in order so Robo reaches the end and reacts to every square.
5. Make (8 min)	Invite pairs to invent their own rule.	Draw five squares, write 'If Robo is on a ___ square, then ___, else ___', swap and predict.

Lesson set-up



Set-up: five floor squares in a row: START, then puddle (grey), plain, puddle, plain. A child plays Robo and moves one square at a time.

Common mistakes to watch for

- ✓ **Thinking both branches run.** Ask: 'Did Robo say Splash AND Step on the same square?'
- ✓ **Thinking the 'if' is checked only once.** Walk it slowly; Robo checks on every square.
- ✓ **Forgetting the else.** Ask what Robo says on a plain square if there is no else.

Different paces

Support: use 3 squares and say the condition aloud on each square.
Stretch: add a third kind of square and ask how to handle three choices (if, else if, else).

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Check for understanding

Two quick questions at the end of the period:

1. On the path plain, puddle, puddle, what does Robo say? (Step, Splash, Splash) 2. Write an if-else rule for 'hot' and 'cold' tiles.

Note in the observation journal: **Beginning** (needs help to trace), **Developing** (traces, cannot yet write a rule), **Secure** (writes a working if-else), **Extending** (handles three choices or explains why the check repeats).

Curriculum link

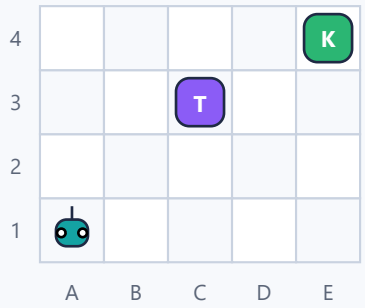
CBSE's Class 4 handbook practises rules with exceptions in *Hide and Seek*. This lesson turns that reasoning into an if-else inside a loop: the same structure children will later write in block coding.

Teacher notes

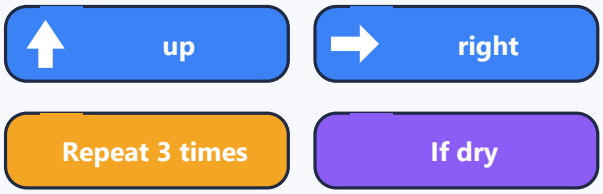
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The classroom kit

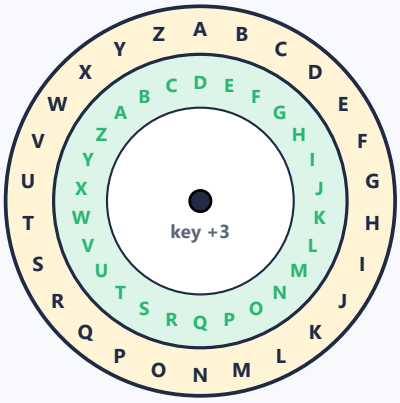
One kit per class section, used all year. Everything is printed, laminated and reusable.



Floor grid mat (6 x 6 ft) for robot routes



Code cards: arrows, repeat, if, when (4 decks)



Cipher wheels for secret codes (20)



Picture and shape cards for patterns and sorting (120)

Item	Quantity per kit
Code card decks (playing-card size, laminated)	4 decks of 54
Cut-out coding blocks (die-cut sheets)	20 sheets
Floor grid mat, washable	1
Cipher wheels	20
Picture and shape cards	120
Robot direction tokens	30
Teacher quick-guide card	1
Sturdy box	1

Illustrations show the design. Final materials will be confirmed after the pilot.

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Assessment and the report card

CBSE asks for continuous, competency-based assessment in CT&AI. The teacher book includes an observation journal using four levels, and a term summary that fits on the report card.

Skill	Beginning	Developing	Secure	Extending
Following steps	Follows short steps with help	Follows and fixes simple steps	Writes steps with loops	Designs and improves own steps
Patterns	Spots simple repeats	Continues a pattern	Finds the core and rule	Makes new patterns and explains them
Breaking problems down	Needs help to start	Breaks familiar jobs into steps	Breaks new problems into steps	Plans steps and their order alone
Logic and rules	Uses one clue	Uses two clues or an if-then rule	Uses several clues and if-else	Spots extra clues and explains

Sample term summary

Area	Level	Teacher comment
Computational thinking (overall)	Secure	Breaks problems into steps and writes loops confidently.
Coding on paper	12 of 14 'build it' tasks	Ready for if-else rules with more than two choices.
Self-check ('How did I do?')	Mostly green	Finds codes easy and clues harder; we practised clue puzzles together.

Children check themselves too

Each chapter ends with **'How did I do?'**, where children colour a face for each 'I can' statement. Teachers can compare this with what they observed, which helps children learn to judge their own progress.

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Answers for the sample chapters

2. Hide and Seek

1. Step, Splash, Splash, Step
2. jump, clap, jump
3. c) 4
4. b) blue
5. a) C2
Build it: Repeat 5 times: move forward; if on a sun tile then say 'Hot!' else say 'Cold!'
Challenge: 3 (tiles 3, 6 and 9)

4. Thousands Around Us

1. 4, 0, 7, 2
2. 3,041; 3,140; 3,401; 3,410
3. b) 3
4. 4, 6, 9
5. 8,510
Build it: Yes: these three comparisons always sort three people
Challenge: No: 3, 2, 1 becomes 2, 3, 1 and then 2, 1, 3, which is not sorted, so the third comparator is needed

9. Equal Groups

1. 15
2. 24
3. a) 3 x 6 = 18
4. 30
5. 2 rows of 3 circles
Build it: Repeat 4 times: [repeat 5 times: plant a seed], start a new row; 20 seeds
Challenge: 6 claps and 3 stamps

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

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A note for parents

Schools can send this page home at the start of the year.

Dear parents,

This year your child will learn **computational thinking** as part of maths, following CBSE's new curriculum for Classes 3-8. Computational thinking means solving problems the way a programmer does: breaking a big job into steps, spotting patterns, keeping what matters and following clear rules. These skills help in maths, reading and everyday life, and they are the base for coding and AI later.

Your child will use the *Think Like a Coder* workbook in class. There are no screens: children solve puzzles, crack secret codes, guide a robot around a grid and write programs with paper code cards.

How you can help at home

- ✓ Ask your child to explain a puzzle to you. Explaining is part of the learning.
- ✓ Play 'robot' at home: your child gives you step-by-step instructions to make a sandwich, and you follow them exactly.
- ✓ Spot patterns together: on clothes, tiles, calendars and number plates.
- ✓ Praise effort and good thinking, not just right answers.

Take-home challenge

Write the numbers 1 to 20. Colour every 3rd number. What pattern do you see? Which number would be the 10th coloured one?

Answer: 3, 6, 9, ... the 10th is 30.

No child's personal data is collected by the workbook. Thank you for your support.

Class teacher _____ **School** _____

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Next steps for your school

1. **Look through this sample**
   Try a chapter with a class or ask a teacher to review it.

2. **Talk to us**
   A short meeting with the principal and coordinator to answer questions.

3. **Start a free pilot**
   Two or three sections, January to March 2027, with an orientation and weekly support.

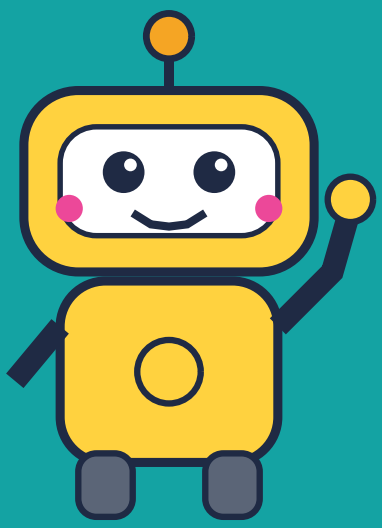
4. **Add it to the 2027-28 book list**
   Most schools finalise their list by the end of January.

Contact

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Notes

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Think Like a Coder

Computational thinking for Classes 3-5, for CBSE schools

Ask us about a free pilot

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